

Classification

ACTL3143 & ACTL5111 Deep Learning for Actuaries
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Lecture Outline

- **Example 1: Binary Classification**
- Example 2: Multiclass Classification
- Summary

Stroke Prediction Data description

1. **id**: unique identifier
2. **gender**: “Male”, “Female” or “Other”
3. **age**: age of the patient
4. **hypertension**: 0 or 1 if the patient has hypertension
5. **heart_disease**: 0 or 1 if the patient has any heart disease
6. **ever_married**: “No” or “Yes”
7. **work_type**: “children”, “Govt_jov”, “Never_worked”, “Private” or “Self-employed”
8. **Residence_type**: “Rural” or “Urban”
9. **avg_glucose_level**: average glucose level in blood
10. **bmi**: body mass index
11. **smoking_status**: “formerly smoked”, “never smoked”, “smokes” or “Unknown”
12. **stroke**: 0 or 1 if the patient had a stroke

Source: Kaggle, **Stroke Prediction Dataset**.

Load up the (pre-)preprocessed data

```

1 PROCESSED_DATA_DIR = Path("stroke/processed")
2
3 X_train = pd.read_csv(PROCESSED_DATA_DIR / "x_train.csv")
4 X_val= pd.read_csv(PROCESSED_DATA_DIR / "x_val.csv")
5 X_test = pd.read_csv(PROCESSED_DATA_DIR / "x_test.csv")
6 y_train = pd.read_csv(PROCESSED_DATA_DIR / "y_train.csv")
7 y_val = pd.read_csv(PROCESSED_DATA_DIR / "y_val.csv")
8 y_test = pd.read_csv(PROCESSED_DATA_DIR / "y_test.csv")
9
10 X_train

```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_Urban
0	0.0	1.0	0.0	1.0	1.0	0.0
1	0.0	1.0	1.0	0.0	1.0	0.0
2	0.0	1.0	1.0	0.0	1.0	0.0
...	...	...	...	...	...	...
3063	1.0	0.0	0.0	1.0	1.0	0.0
3064	1.0	0.0	0.0	1.0	1.0	0.0
3065	0.0	1.0	1.0	0.0	0.0	1.0

3066 rows × 20 columns

Target variable

```
1 y_train
```

	stroke
0	0
1	0
2	0
...	...
3063	0
3064	0
3065	0

3066 rows × 1 columns

```
1 import numpy as np
2 classes, counts = np.unique(y_train.values.ravel(), return_counts=True)
3 print("Classes:", classes)
4 print("Counts:", counts)
```

Classes: [0 1]

Counts: [2909 157]

Setup a binary classification model

```

1 def create_model(seed=42):
2     random.seed(seed)
3     model = Sequential()
4     model.add(Input(X_train.shape[1:]))
5     model.add(Dense(32, "leaky_relu"))
6     model.add(Dense(16, "leaky_relu"))
7     model.add(Dense(1, "sigmoid"))
8     return model

```

```

1 model = create_model()
2 model.summary()

```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 32)	672
dense_1 (Dense)	(None, 16)	528
dense_2 (Dense)	(None, 1)	17

Total params: 1,217 (4.75 KB)

Trainable params: 1,217 (4.75 KB)

Non-trainable params: 0 (0.00 B)

Fit the model

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy")  
3 model.fit(X_train, y_train, epochs=5, verbose=2)
```

Epoch 1/5

96/96 - 0s - 3ms/step - loss: 0.2734

Epoch 2/5

96/96 - 0s - 2ms/step - loss: 0.1753

Epoch 3/5

96/96 - 0s - 2ms/step - loss: 0.1665

Epoch 4/5

96/96 - 0s - 3ms/step - loss: 0.1619

Epoch 5/5

96/96 - 0s - 2ms/step - loss: 0.1595

<keras.src.callbacks.history.History at 0x12359acf0>

Track accuracy as the model trains

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])  
3 model.fit(X_train, y_train, epochs=5, verbose=2)
```

Epoch 1/5

96/96 - 0s - 2ms/step - accuracy: 0.9204 - loss: 0.2711

Epoch 2/5

96/96 - 0s - 3ms/step - accuracy: 0.9488 - loss: 0.1766

Epoch 3/5

96/96 - 0s - 3ms/step - accuracy: 0.9488 - loss: 0.1667

Epoch 4/5

96/96 - 0s - 3ms/step - accuracy: 0.9488 - loss: 0.1623

Epoch 5/5

96/96 - 0s - 3ms/step - accuracy: 0.9488 - loss: 0.1595

<keras.src.callbacks.history.History at 0x12186a0d0>

Run a long fit

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])  
3 %time hist = model.fit(X_train, y_train, epochs=500, validation_data=(X_val, y_val), verb
```

CPU times: user 2min 31s, sys: 3.46 s, total: 2min 34s

Wall time: 2min 31s

Add early stopping

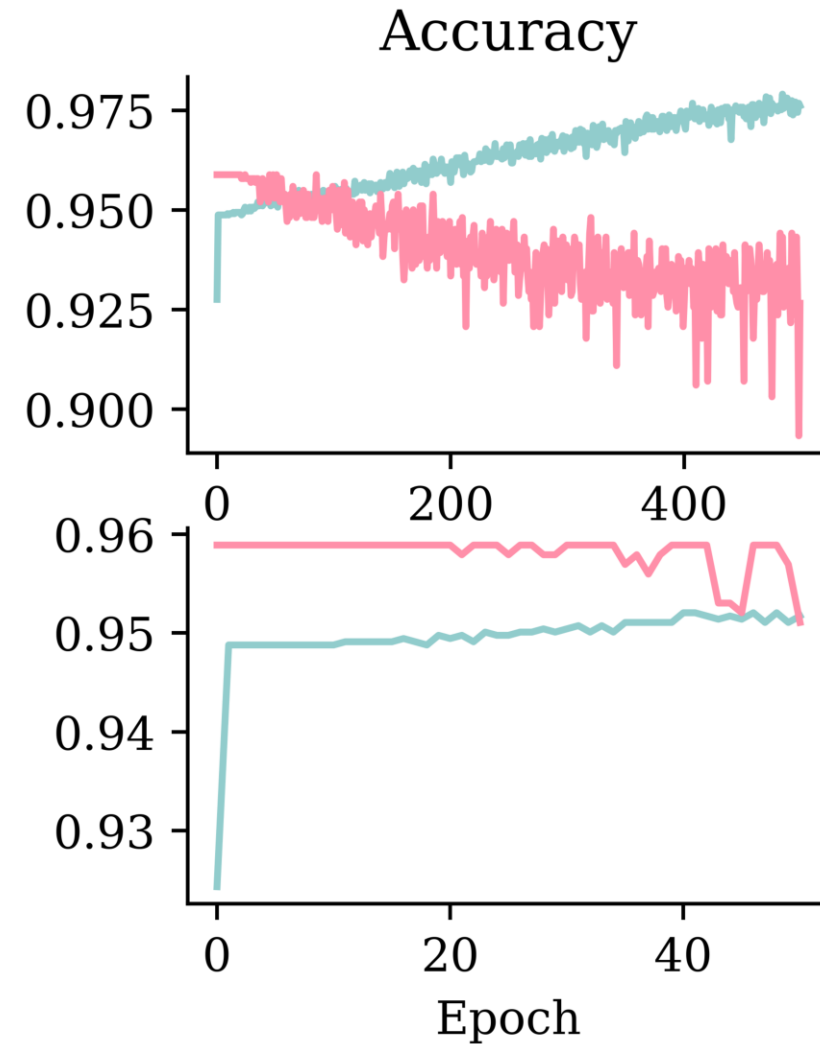
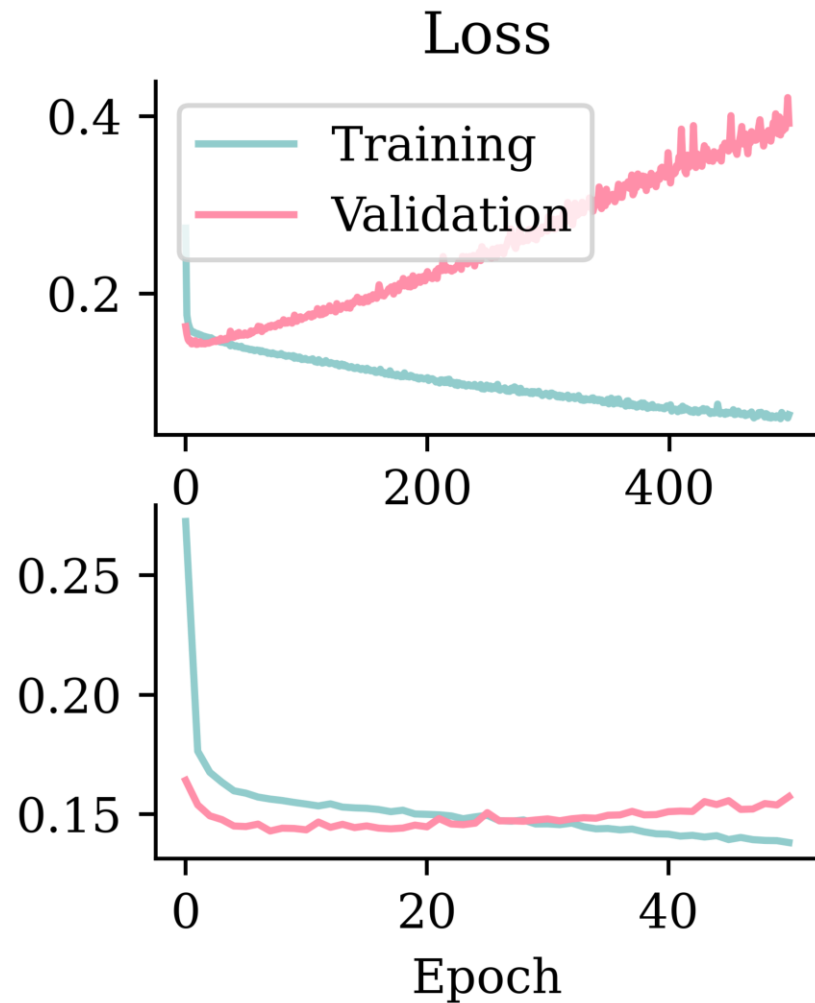
```
1 model = create_model()
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])
3 es = EarlyStopping(restore_best_weights=True, patience=50, monitor="val_accuracy")
4 %time hist_es = model.fit(X_train, y_train, epochs=500, validation_data=(X_val, y_val), c
5 print(f"Stopped after {len(hist_es.history['loss'])} epochs.")
```

CPU times: user 15.4 s, sys: 347 ms, total: 15.7 s

Wall time: 15.4 s

Stopped after 51 epochs.

Fitting metrics



Add metrics, compile, and fit

```
1 model = create_model()
2
3 pr_auc = keras.metrics.AUC(curve="PR", name="pr_auc")
4 model.compile(optimizer="adam", loss="binary_crossentropy",
5               metrics=[pr_auc, "accuracy", "auc"])
6
7 es = EarlyStopping(patience=50, restore_best_weights=True,
8                   monitor="val_pr_auc", verbose=1)
9 model.fit(X_train, y_train, callbacks=[es], epochs=1_000, verbose=0,
10         validation_data=(X_val, y_val));
```

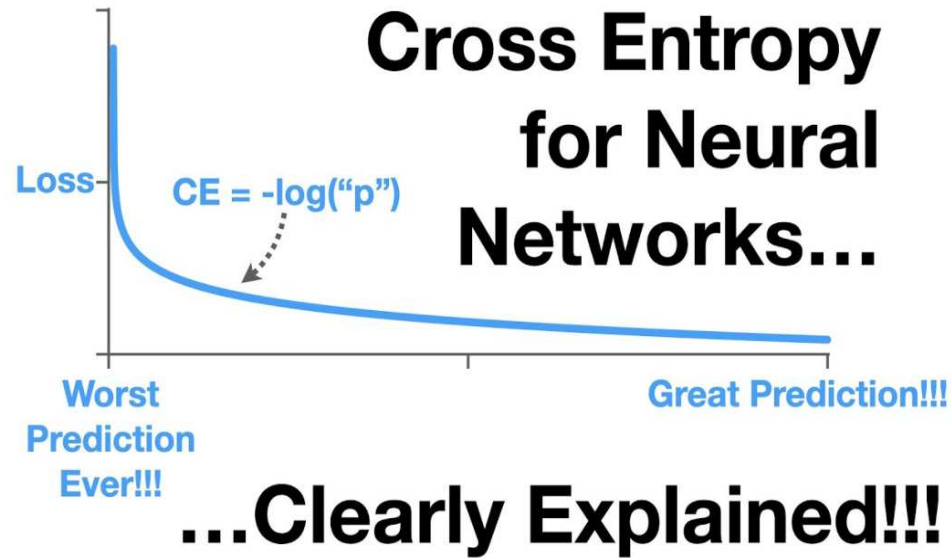
Epoch 81: early stopping

Restoring model weights from the end of the best epoch: 31.

```
1 model.evaluate(X_val, y_val, verbose=0)
```

```
[0.14898666739463806,
0.12857568264007568,
0.9569471478462219,
0.8119411468505859]
```

Cross-entropy loss: ELI5

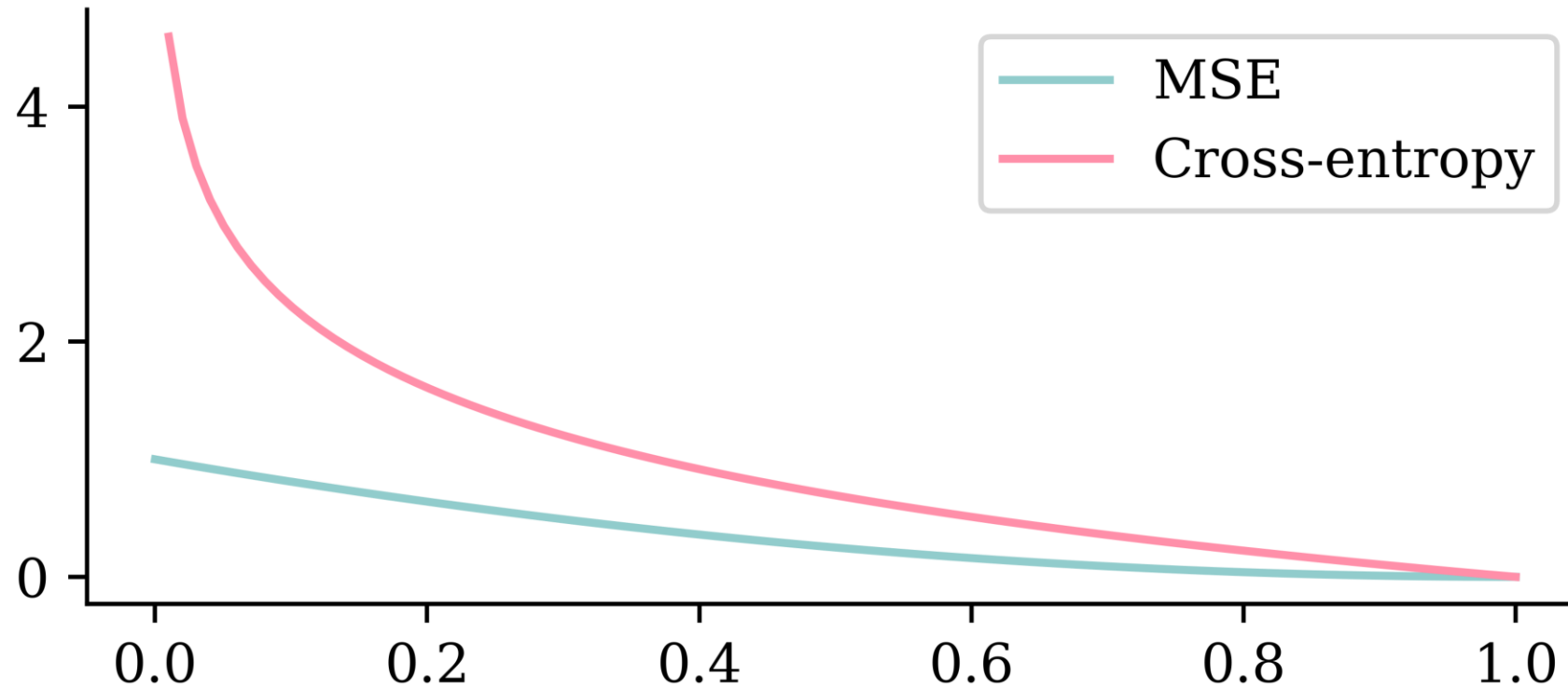


Cross Entropy for Neural Networks...



Why use cross-entropy loss?

```
1 p = np.linspace(0, 1, 100)
2 plt.plot(p, (1 - p) ** 2)
3 plt.plot(p, -np.log(p))
4 plt.legend(["MSE", "Cross-entropy"]);
```



Overweight the minority class

```

1 model = create_model()
2
3 pr_auc = keras.metrics.AUC(curve="PR", name="pr_auc")
4 model.compile(optimizer="adam", loss="binary_crossentropy",
5               metrics=[pr_auc, "accuracy", "auc"])
6
7 es = EarlyStopping(patience=50, restore_best_weights=True,
8                   monitor="val_pr_auc", verbose=1)
9 model.fit(X_train, y_train.to_numpy(), callbacks=[es], epochs=1_000, verbose=0,
10         validation_data=(X_val, y_val), class_weight={0: 1, 1: 10});

```

Epoch 64: early stopping

Restoring model weights from the end of the best epoch: 14.

```
1 model.evaluate(X_val, y_val, verbose=0)
```

```

[0.3523019552230835,
 0.13380154967308044,
 0.7896282076835632,
 0.8259596824645996]

```

```
1 model.evaluate(X_test, y_test, verbose=0)
```

```

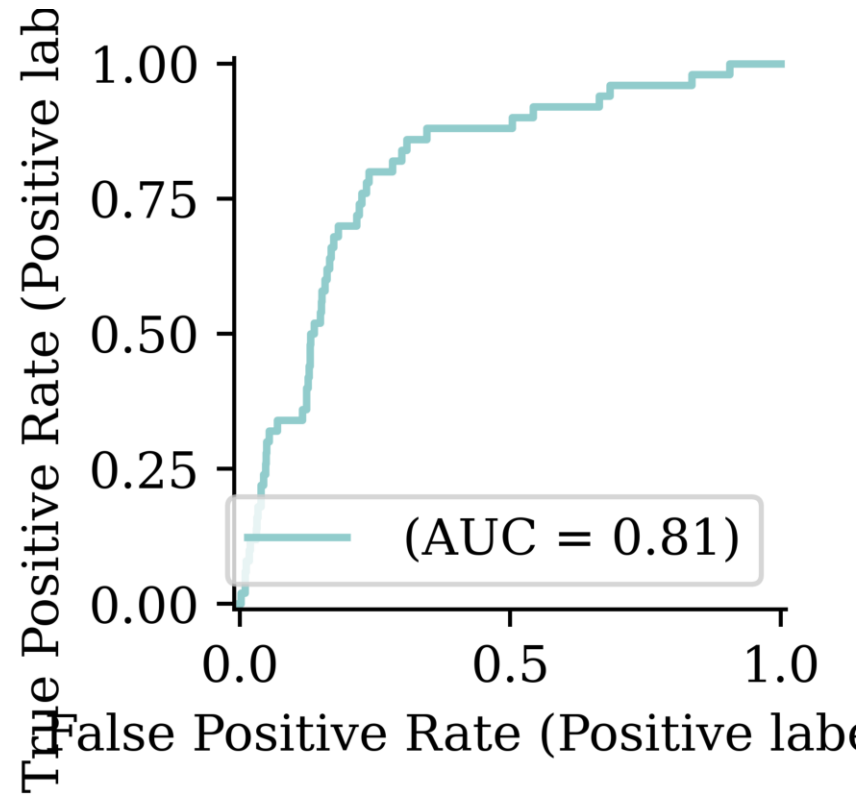
[0.36996063590049744,
 0.15842117369174957,
 0.7954990267753601,
 0.8060390949249268]

```

Classification Metrics

```
1 from sklearn.metrics import confusion_matrix, RocCurveDisplay, PrecisionRecallDisplay
2 y_pred = model.predict(X_test, verbose=0)
```

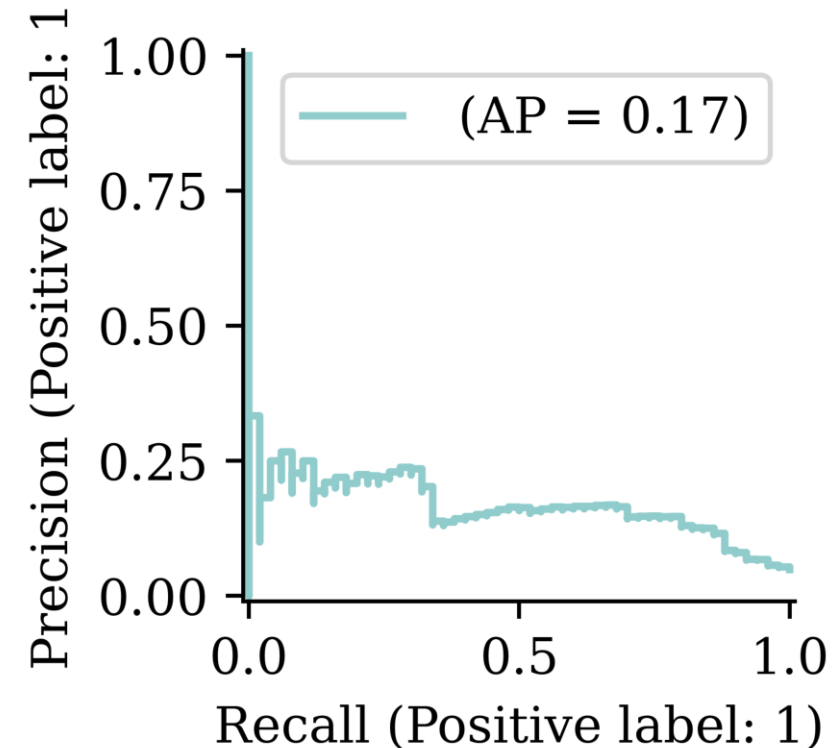
```
1 RocCurveDisplay.from_predictions(y_test, y_pred, name="");
```



```
1 y_pred_stroke = y_pred > 0.5
2 confusion_matrix(y_test, y_pred_stroke)
```

```
array([[778, 194],
       [ 15, 251]])
```

```
1 PrecisionRecallDisplay.from_predictions(y_test, y_pred, name="");
```



```
1 y_pred_stroke = y_pred > 0.3
2 confusion_matrix(y_test, y_pred_stroke)
```

```
array([[647, 325],
       [  7, 211]])
```


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Iris dataset

```

1 from sklearn.datasets import load_iris
2 iris = load_iris()
3 names = ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]
4 features = pd.DataFrame(iris.data, columns=names)
5 features

```

	SepalLength	SepalWidth	PetalLength	PetalWidth
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
...	...	...	...	...
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

Target variable

```
1 iris.target_names
```

```
array(['setosa', 'versicolor', 'virginica'],  
      dtype='<U10')
```

```
1 iris.target[:8]
```

```
array([0, 0, 0, 0, 0, 0, 0, 0])
```

```
1 target = iris.target  
2 target = target.reshape(-1, 1)  
3 target[:8]
```

```
array([[0],  
       [0],  
       [0],  
       [0],  
       [0],  
       [0],  
       [0]])
```

```
1 classes, counts = np.unique(  
2     target,  
3     return_counts=True  
4 )  
5 print(classes)  
6 print(counts)
```

```
[0 1 2]  
[50 50 50]
```

```
1 iris.target_names[  
2     target[[0, 30, 60]]  
3 ]
```

```
array(['setosa',  
       'setosa',  
       'versicolor'], dtype='<U10')
```

Split the data into train and test

```
1 X_train, X_test, y_train, y_test = train_test_split(features, target, random_state=24)
2 X_train
```

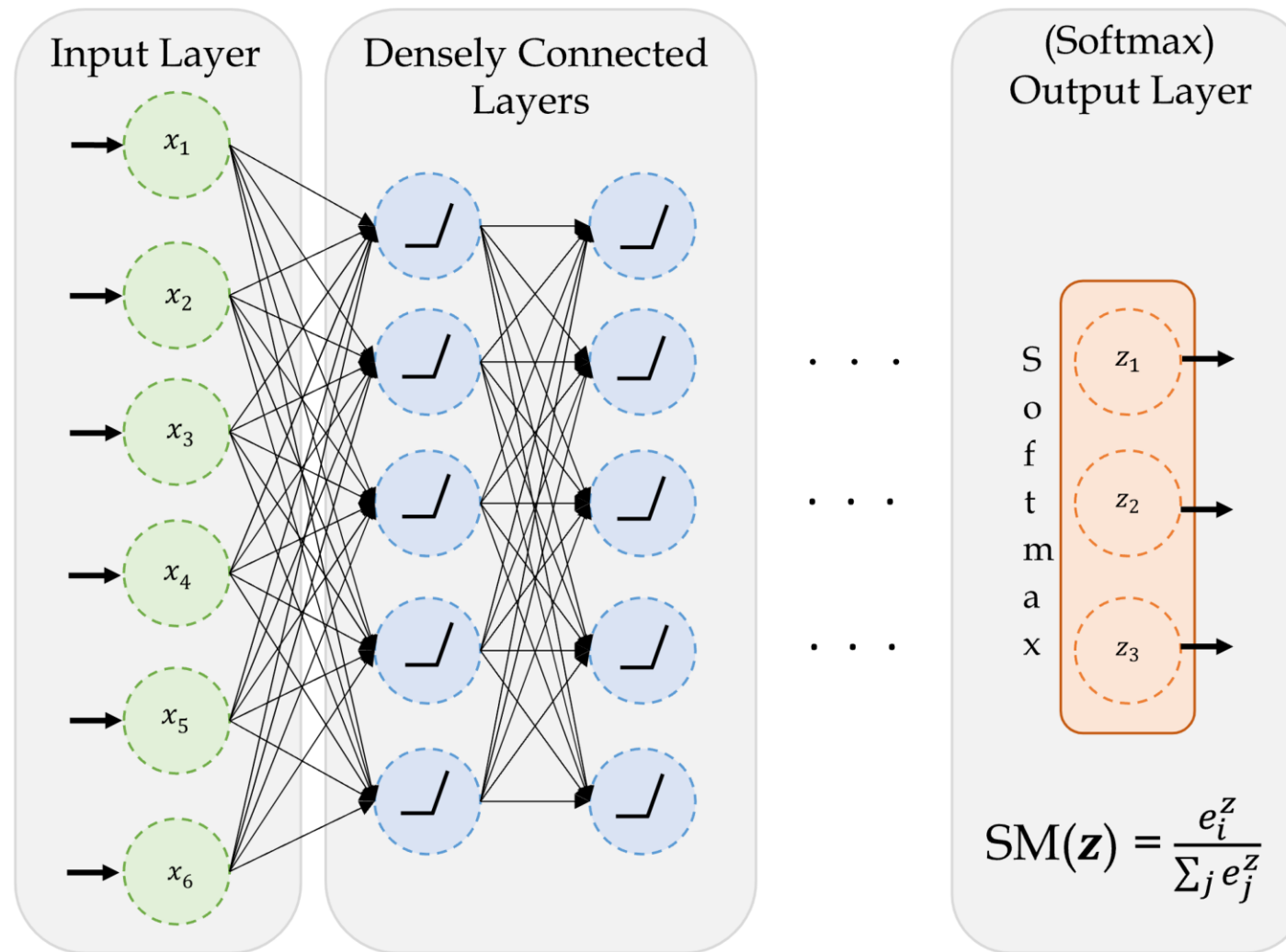
	SepalLength	SepalWidth	PetalLength	PetalWidth
53	5.5	2.3	4.0	1.3
58	6.6	2.9	4.6	1.3
95	5.7	3.0	4.2	1.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
87	6.3	2.3	4.4	1.3
131	7.9	3.8	6.4	2.0

112 rows $\times$ 4 columns

```
1 X_test.shape, y_test.shape
```

((38, 4), (38, 1))

A basic classifier network



A basic network for classifying into three categories.

Source: Marcus Lautier (2022).

Create a classifier model

```
1 NUM_FEATURES = len(features.columns)
2 NUM_CATS = len(np.unique(target))
3
4 print("Number of features:", NUM_FEATURES)
5 print("Number of categories:", NUM_CATS)
```

Number of features: 4

Number of categories: 3

Make a function to return a Keras model:

```
1 def build_model(seed=42):
2     random.seed(seed)
3     return Sequential([
4         Dense(30, activation="relu"),
5         Dense(NUM_CATS, activation="softmax")
6     ])
```

Fit the model

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy")  
3  
4 model.fit(X_train, y_train, epochs=5, verbose=2);
```

Epoch 1/5

4/4 - 0s - 2ms/step - loss: 1.3920

Epoch 2/5

4/4 - 0s - 2ms/step - loss: 1.2912

Epoch 3/5

4/4 - 0s - 2ms/step - loss: 1.2196

Epoch 4/5

4/4 - 0s - 2ms/step - loss: 1.1576

Epoch 5/5

4/4 - 0s - 2ms/step - loss: 1.1084

Track accuracy as the model trains

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy", metrics=["accuracy"])  
3 model.fit(X_train, y_train, epochs=5, verbose=2);
```

Epoch 1/5

4/4 - 0s - 2ms/step - accuracy: 0.2857 - loss: 1.3930

Epoch 2/5

4/4 - 0s - 2ms/step - accuracy: 0.2857 - loss: 1.2970

Epoch 3/5

4/4 - 0s - 2ms/step - accuracy: 0.2857 - loss: 1.2203

Epoch 4/5

4/4 - 0s - 2ms/step - accuracy: 0.2946 - loss: 1.1596

Epoch 5/5

4/4 - 0s - 2ms/step - accuracy: 0.3393 - loss: 1.1067

Run a long fit

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy", \  
3     metrics=["accuracy"])  
4 %time hist = model.fit(X_train, y_train, epochs=500, \  
5     validation_split=0.25, verbose=False)
```

CPU times: user 5.19 s, sys: 495 ms, total: 5.68 s

Wall time: 5.36 s

Evaluation now returns both *loss* and *accuracy*.

```
1 model.evaluate(X_test, y_test, verbose=False)
```

[0.08639740198850632, 0.9736841917037964]

Add early stopping

```

1 model_es = build_model()
2 model_es.compile("adam", "sparse_categorical_crossentropy", \
3                 metrics=["accuracy"])
4
5 es = EarlyStopping(restore_best_weights=True, patience=50,
6                   monitor="val_accuracy")
7 %time hist_es = model_es.fit(X_train, y_train, epochs=500, \
8                             validation_split=0.25, callbacks=[es], verbose=False);
9
10 print(f"Stopped after {len(hist_es.history['loss'])} epochs.")

```

CPU times: user 748 ms, sys: 69 ms, total: 817 ms

Wall time: 772 ms

Stopped after 70 epochs.

Evaluation on test set:

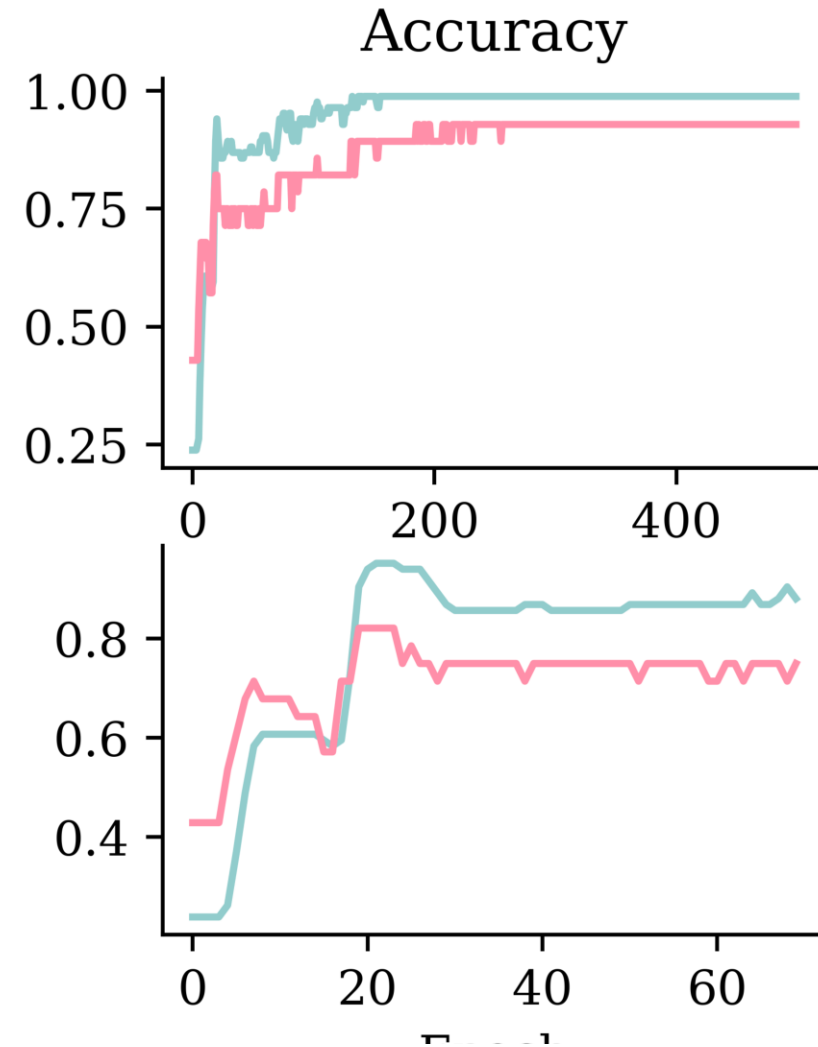
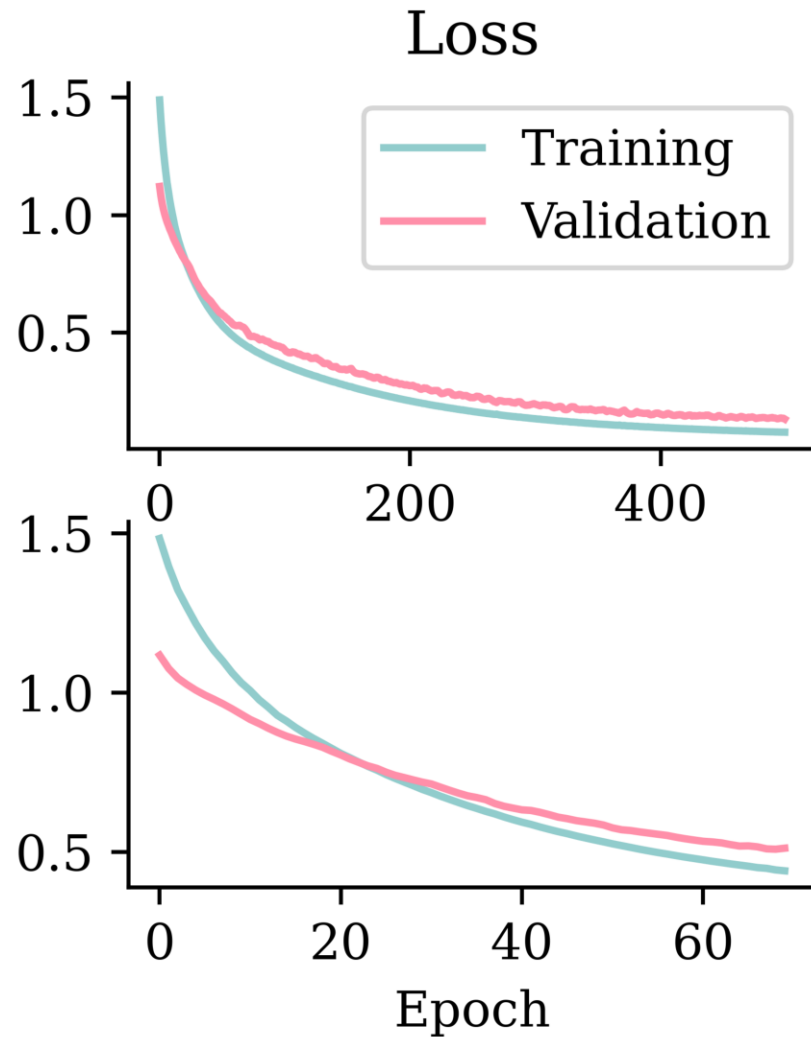
```

1 model_es.evaluate(X_test, y_test, verbose=False)

```

[0.8077937960624695, 0.9210526347160339]

Fitting metrics



What is the softmax activation?

It creates a “probability” vector: $\text{Softmax}(\mathbf{x}) = \frac{e_i^x}{\sum_j e_j^x}$.

In NumPy:

```
1 out = np.array([5, -1, 6])  
2 (np.exp(out) / np.exp(out).sum()).round(3)
```

```
array([0.27, 0.  , 0.73])
```

In Keras:

```
1 out = keras.ops.convert_to_tensor([[5.0, -1.0, 6.0]])  
2 keras.ops.round(keras.ops.softmax(out), 3)
```

```
tensor([[0.2690, 0.0010, 0.7310]])
```

Prediction using classifiers

```
1 y_test[:4]
```

```
array([[2],
       [2],
       [1],
       [1]])
```

```
1 y_pred = model.predict(X_test.head(4), verbose=0)
2 y_pred
```

```
array([[2.02e-06, 7.64e-02, 9.24e-01],
       [1.86e-07, 1.62e-03, 9.98e-01],
       [1.44e-02, 9.76e-01, 1.00e-02],
       [2.80e-03, 8.50e-01, 1.48e-01]], dtype=float32)
```

```
1 # Add 'keepdims=True' to get a column vector.
2 np.argmax(y_pred, axis=1)
```

```
array([2, 2, 1, 1])
```

```
1 iris.target_names[np.argmax(y_pred, axis=1)]
```

```
array(['virginica', 'virginica', 'versicolor', 'versicolor'], dtype='<U10')
```

Lecture Outline

- Example 1: Binary Classification
- Example 2: Multiclass Classification
- **Summary**

Classification models in Keras

If the number of classes is c , then:

Target	Output Layer	Loss Function
Binary ($c = 2$)	1 neuron with <code>sigmoid</code> activation	Binary Cross-Entropy
Multi-class ($c > 2$)	c neurons with <code>softmax</code> activation	Categorical Cross-Entropy

Optionally output logits

If the number of classes is c , then:

Target	Output Layer	Loss Function
Binary ($c = 2$)	1 neuron with <code>linear</code> activation	Binary Cross-Entropy (<code>from_logits=True</code>)
Multi-class ($c > 2$)	c neurons with <code>linear</code> activation	Categorical Cross-Entropy (<code>from_logits=True</code>)

Code examples

Binary

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(1, activation="sigmoid")
4 ])
5 model.compile(loss="binary_crossentropy")
```

Binary (logits)

```
1 from keras.losses import BinaryCrossentropy
2 model = Sequential([
3     # Skipping the earlier layers
4     Dense(1, activation="linear")
5 ])
6 loss = BinaryCrossentropy(from_logits=True)
7 model.compile(loss=loss)
```

Multi-class

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(n_classes, activation="softmax")
4 ])
5 model.compile(loss="sparse_categorical_crossentropy")
```

Multi-class (logits)

```
1 from keras.losses import SparseCategoricalCrossentropy
2
3 model = Sequential([
4     # Skipping the earlier layers
5     Dense(n_classes, activation="linear")
6 ])
7 loss = SparseCategoricalCrossentropy(from_logits=True)
8 model.compile(loss=loss)
```

Package Versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch")
```

Python implementation: CPython

Python version : 3.14.3

IPython version : 9.13.0

keras : 3.14.1

matplotlib: 3.10.9

numpy : 2.4.4

pandas : 3.0.2

seaborn : 0.13.2

scipy : 1.17.1

torch : 2.11.0

Glossary

- accuracy
- classification problem
- confusion matrix
- cross-entropy loss
- metrics
- sigmoid activation function
- softmax activation